An upper bound on the Berry–Esseen constant for independent summands

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Abstract

For every finite sum of independent centered real random variables with positive total variance and finite third absolute moments, the Kolmogorov distance from the standard normal law after normalization to variance one is at most U_*L. Here L is the normalized sum of third absolute moments and $U_* < 0.44988$. The proof compares characteristic functions with those of symmetric two-point variables having the same individual variances. Their real characteristic-function product retains a signed contribution to the smoothing integral. Third-moment excess controls the comparison error; the largest summand variance and its third moment sharpen that control. Analytic estimates cover small and large L, and a finite interval calculation covers the remaining moment domain. The accompanying certificate gives the complete partitions, numerical enclosures, and arithmetic sources.

1 The problem and the upper bound

Let $X_1, \dots, X_n$ be a finite independent family of centered real random variables. Assume

$$0 < V = \sum_{j=1}^{n} \mathbb{E}X_j^2 < \infty, \qquad \sum_{j=1}^{n} \mathbb{E}|X_j|^3 < \infty.$$

Write Φ and ϕ for the standard normal distribution function and density, and set

$$S = \frac{\sum_j X_j}{\sqrt{V}}, \qquad L = \frac{\sum_j \mathbb{E}|X_j|^3}{V^{3/2}}, \qquad \Delta = \sup_{x\in\mathbb{R}} |\mathbb{P}(S \le x) - \Phi(x)|. \tag{1.1}$$

Thus $C_{\mathrm{ind}} = \sup \Delta/L$, over all such arrays, is the unrestricted independent Berry–Esseen constant. Esseen's lower bound is

$$C_{\mathrm{E}} = \frac{\sqrt{10}+3}{6\sqrt{2\pi}} = 0.40973218370239634299\ldots \le C_{\mathrm{ind}}; \tag{1.2}$$

see Esseen [1] and the discussion in Shevtsova [12]. The conjectured equality $C_{\mathrm{ind}} = C_{\mathrm{E}}$ remains open.

Theorem 1.1. *For the arrays in* (1.1)*,*

$$\begin{gathered}\Delta \le U_* L, \qquad C_{\mathrm{E}} \le C_{\mathrm{ind}} \le U_* < 0.44988, \\ U_* = \frac{1013039690742047}{2251799813685248}.\end{gathered} \tag{1.3}$$

For comparison, Shevtsova [11] obtained the upper bound 0.5583 for general independent summands. The interval calculation proving the new upper bound is given in Section 10.

The proof refines the zero-bias and Fourier estimates of Tyurin [13] by retaining third-moment excess and logarithmic corrections to characteristic-function products. It uses Prawitz smoothing and comparison with a sum of symmetric two-point variables having the same variances, an approximation studied by Mattner and Shevtsova [6, Theorem 1.2 and Example 1.4]. All estimates use only the individual variances and third absolute moments.

The main difficulty is to control the characteristic-function error uniformly over arrays with unequal variances. The classical modulus estimates do not exploit the vanishing comparison error at zero third-moment excess or the sign of the cosine-product displacement. Write f_S for the sum's characteristic function, C_v for the cosine product, and g for the Gaussian characteristic function. We split $f_S - g = (f_S - C_v) + (C_v - g)$. The first term is bounded by an error radius. Before the first positive zero of a cosine factor, the second is real and nonpositive, so it contributes with its sign in Prawitz's smoothing integral. The resulting estimate depends on four moment parameters, whose feasible domain admits a finite certified cover.

1.1 Normalization and moment parameters

Replace each X_j by $X_j/\sqrt{V}$, reuse the notation X_j, and discard summands of variance zero. Then $S = \sum_j X_j$ and the total variance is one. Put

$$v_j = \mathbb{E}X_j^2, \qquad b_j = \mathbb{E}|X_j|^3, \qquad \tau = \sum_j v_j^{3/2}, \qquad \sum_j v_j = 1, \quad \sum_j b_j = L.$$

The differences $b_j - v_j^{3/2}$ are nonnegative and sum to $L - \tau$. They vanish for centered symmetric two-point summands. Relabel the summands so that $v_1 = \max_j v_j$, and put $d = v_1$ and $b = b_1$. The moment parameters satisfy

$$b \ge d^{3/2}, \qquad d^{3/2} \le \tau \le \min\{L - b + d^{3/2}, \sqrt{d}\}, \qquad \tau \le L. \tag{1.4}$$

Indeed $b_j \ge v_j^{3/2}$ and $v_j \le d$ imply $\tau \le \sqrt{d}\sum_j v_j = \sqrt{d}$. In particular $\tau^2 \le d \le \tau^{2/3}$. For the remaining summands, write

$$w = 1 - d, \qquad B_R = L - b, \qquad \tau_R = \tau - d^{3/2}, \qquad a_R = \min\{d, w, B_R^{2/3}, \tau_R^{2/3}\}. \tag{1.5}$$

For $j \ge 2$, we have $v_j \le d$, $v_j \le w$, $v_j^{3/2} \le b_j \le B_R$, and $v_j^{3/2} \le \tau_R$. Thus $v_j \le a_R$. Every realizable tuple satisfies these constraints; maximizing over the possibly larger set they define gives an upper bound.

The product and zero-bias estimates in Sections 2–5 bound $|f_S(t) - e^{-t^2/2}|$. Section 6 uses the largest-variance summand separately. The comparison with $\prod_i \cos(t\sqrt{v_i})$ in Section 7 preserves a signed real error. Smoothing then bounds Δ/L, and Section 10 evaluates that bound on a complete cover of the moment domain.

2 Bounds for characteristic-function products

For a random variable X, let $f_X(t) = \mathbb{E}e^{itX}$, and write $f_j = f_{X_j}$ for the summands. Since $f_X(-t) = \overline{f_X(t)}$, it suffices to bound its modulus for $t \ge 0$. For a nonempty index set J, put

$$V_0 = \sum_{j\in J} v_j, \qquad B = \sum_{j\in J} b_j, \qquad \tau_0 = \sum_{j\in J} v_j^{3/2},$$

and let a satisfy $v_j \le a$ for every $j \in J$. We need bounds for the product over J and for the product after any one index is omitted. Products over the empty set equal one. The products with one factor omitted arise from the zero-bias replacement identity (6.1), which gives

$$|f_S(t) - f_{S^*}(t)| \le \sum_i v_i |f_{X_i}(t) - f_{X_i^*}(t)| \prod_{j \ne i} |f_j(t)|.$$

2.1 Cubic and trigonometric bounds

Define

$$\kappa_* = \sup_{x>0} \frac{\cos x - 1 + x^2/2}{x^3}, \qquad \kappa = \frac{49581}{500000} = 0.099162. \tag{2.1}$$

Let θ be the maximizer in (2.1). For $V_0, E_0 > 0$, $t \ge 0$, and $u = E_0 t / V_0$, put

$$\psi(t; V_0, E_0) = \begin{cases} V_0 t^2/2 - \kappa_* E_0 t^3, & u < \theta, \\ V_0^3 E_0^{-2}(1 - \cos u), & \theta \le u \le 2\pi, \\ 0, & u > 2\pi. \end{cases} \tag{2.2}$$

The characteristic-function inequalities of Prawitz [10], as stated in Tyurin [13, Lemma 1], yield, with $v = \mathbb{E}X^2 > 0$ and $b = \mathbb{E}|X|^3 \ge v^{3/2}$ in the second formula,

$$\Big| \prod_j f_{X_j}(t) \Big| \le \exp\{-\psi(t; V_0, B + \tau_0)\}, \tag{2.3}$$

$$|f_X(t)| \le A(t; v, b) := \sqrt{1 - 2\psi(t; v, b + v^{3/2})}. \tag{2.4}$$

For fixed parameters, ψ first increases and then decreases with t, and is zero after the cosine branch. On the cubic branch, $\partial_t \psi = t(V_0 - 3\kappa_* E_0 t)$; on the cosine branch, $\partial_t \psi = (V_0^2/E_0)\sin u \le 0$. Thus the upper bounds in (2.3)–(2.4) attain their maxima on a frequency interval at an endpoint. On the cosine branch,

$$\partial_{V_0} \psi = \frac{V_0^2}{E_0^2}\{3(1 - \cos u) - u \sin u\} \ge 0,$$
$$\partial_{E_0} \psi = \frac{V_0^3}{E_0^3}\{u \sin u - 2(1 - \cos u)\} \le 0.$$

The same signs hold on the cubic branch. The maximizing equation for θ makes the formulas agree at the joins. Consequently, as functions of the independent arguments of ψ, the upper bounds decrease with V_0 and increase with E_0 on these domains. For a moment box, E_0 is bounded separately; in (2.4) it includes the term $v^{3/2}$.

For $r(x) = (\cos x - 1 + x^2/2)/x^3$, differentiation gives

$$r'(x) = -\frac{h(x)}{2x^4}, \quad h(x) = x^2 + 2x \sin x + 6(\cos x - 1), \quad h'(x) = 4\cos(x/2)\{x\cos(x/2) - 2\sin(x/2)\}.$$

The expression in braces decreases from zero on $(0, 2\pi)$. Consequently h decreases on $(0, \pi)$ and increases on $(\pi, 2\pi)$, with $h(\pi) = \pi^2 - 12 < 0$ and $h(2\pi) = 4\pi^2 > 0$. Thus h has exactly one zero in $(\pi, 2\pi)$. For $x \ge 2\pi$, $r(x) \le 1/(4\pi) < 3/32 \le r(4)$. The zero is therefore the unique global maximizer of r. The scalar checker isolates it in $(3.99, 4)$ and encloses $\kappa_* < 0.099161913515 < \kappa$ with Arb, Johansson's library for rigorous ball arithmetic [4].

The cubic expression $V_0 t^2/2 - \kappa E_0 t^3$ is a lower bound for (2.2) at every frequency. The numerical cosine lower bound is used only when the entire argument interval lies in $[4, 2\pi]$, which is contained in the cosine branch since $\theta < 4$.

2.2 A logarithmic correction

For each summand indexed by J, define

$$h_j = \frac{v_j t^2}{2} - \kappa(b_j + v_j^{3/2})t^3, \qquad q_j = \max(h_j, 0), \qquad H = \frac{V_0 t^2}{2} - \kappa(B + \tau_0)t^3.$$

Since $b_j \geq v_j^{3/2}$,

$$0 \leq q_j \leq m(t,a) := \max_{0 \leq x \leq t\sqrt{a}} (x^2/2 - 2\kappa x^3) \leq \frac{1}{216\kappa^2} < \frac{1}{2}. \tag{2.5}$$

The maximum in m occurs at $x = \min(t\sqrt{a}, 1/(6\kappa))$. The single-summand bound gives $|f_{X_j}(t)| \leq \sqrt{1-2q_j}$. Applying $-\frac{1}{2}\log(1-2q) \geq q + q^2$ and using $\sum_j q_j \geq \sum_j h_j = H$ gives

$$\prod_j |f_{X_j}(t)| \leq \exp\left\{-H - \sum_j q_j^2\right\}. \tag{2.6}$$

Lemma 2.1. *Let $A_0 = t^2/2 - 2\kappa\sqrt{a}\,t^3$. When $A_0 \geq 0$,*

$$\sum_j q_j^2 \geq Q := \left(\max\left\{0, A_0 \frac{\tau_0}{\sqrt{V_0}} - \kappa t^3(B - \tau_0)\right\}\right)^2. \tag{2.7}$$

For $A_0 < 0$, take $Q = 0$.

Proof. Since $v_j \leq a$ and $q_j \geq h_j$,

$$q_j + \kappa t^3(b_j - v_j^{3/2}) \geq v_j(t^2/2 - 2\kappa\sqrt{v_j}\,t^3) \geq A_0 v_j.$$

When $A_0 \geq 0$, taking ℓ_2 norms and applying the triangle inequality gives

$$\|(q_j)\|_2 \geq A_0\|(v_j)\|_2 - \kappa t^3(B - \tau_0).$$

Cauchy–Schwarz gives $\tau_0 \leq \sqrt{V_0}\|(v_j)\|_2$. Taking the maximum of the right side and zero, then squaring, proves (2.7). □

After omitting index i, the linear sum is $H - h_i \geq H - m$. Also $\sum_{j \neq i} q_j^2 \geq \max(0, Q - m^2)$. Hence

$$\begin{aligned} F_0(t; V_0, B, a, \tau_0) &= \min\{1, e^{-H-Q}\}, \\ D_0(t; V_0, B, a, \tau_0) &= \min\{1, e^{-H+m-\max(0,Q-m^2)}\}. \end{aligned} \tag{2.8}$$

Here D_0 bounds the product after any one index is omitted. We may also omit the quadratic correction in either expression. When several upper bounds apply, we take their pointwise minimum. For the empty index set, set $F_0 = D_0 = 1$.

To bound (2.8) when τ_0 ranges over an interval $[\ell, h]$, suppose first that $A_0 \geq 0$ and $t > 0$. Put

$$k = \kappa t^3, \qquad a_0 = A_0/\sqrt{V_0} + k, \qquad c_0 = kB.$$

Apart from terms independent of τ_0, either exponent has the form

$$k\tau_0 - \max\{0, (a_0\tau_0 - c_0)_+^2 - m_0^2\}, \qquad z_+ = \max(z, 0),$$

where $m_0 = 0$ for F_0 and $m_0 = m$ for D_0. Its derivative is k until $a_0\tau_0 - c_0 = m_0$, and then $k - 2a_0(a_0\tau_0 - c_0)$. It is maximized on $[\ell, h]$ at

$$\min\{h, \max\{\ell, \tau_*\}\}, \qquad \tau_* = \frac{c_0 + \max\{m_0, k/(2a_0)\}}{a_0}.$$

When $A_0 < 0$, $Q = 0$ and the exponent is increasing in τ_0. At $t = 0$ both product bounds equal one.

2.3 The full logarithmic correction

We now bound the terms beyond the linear term in $-\frac{1}{2}\log(1-2q)$, using the same moment parameters.

For a subarray of total variance V, third-moment sum B, $\tau_0 = \sum_j v_j^{3/2}$, and individual variances at most a, write

$$k = \kappa t^3, \quad \epsilon_j = b_j - v_j^{3/2}, \quad r_j = v_j t^2/2 - 2k v_j^{3/2}, \quad q_j = (r_j - k\epsilon_j)_+, \quad H = Vt^2/2 - k(B+\tau_0).$$

The elementary characteristic-function inequality gives $|f_j(t)| \le \sqrt{1-2q_j}$ and $0 \le q_j \le m(t,a) < 1/2$. Put

$$g(q) = -\tfrac{1}{2}\log(1-2q) - q, \qquad 0 \le q < 1/2.$$

For $\lambda \ge 0$ and $0 \le r < 1/2$, define

$$\varphi_\lambda(r) = \inf_{0\le q\le r}\{g(q)+\lambda(r-q)\} = \begin{cases} g(r), & r \le q_\lambda, \\ \lambda r - c_\lambda, & r \ge q_\lambda, \end{cases} \quad q_\lambda = \frac{\lambda}{2(1+\lambda)}, \quad c_\lambda = \frac{\lambda - \log(1+\lambda)}{2}. \tag{2.9}$$

The formula follows by minimizing the convex function of q, whose derivative is $2q/(1-2q) - \lambda$.

Lemma 2.2 (A lower bound for the logarithmic correction)**.** *Let $A = 1/(32\kappa^2)$ and $r(z) = Az^2(1-z)$. Suppose $\alpha, \beta \ge 0$ and $0 < M \le 1$ satisfy*

$$\varphi_\lambda(r(z)) \ge z^2(\alpha z - \beta) \qquad (0 \le z \le M). \tag{2.10}$$

If $4\kappa t\sqrt{a} \le M$, set

$$S_\lambda = \frac{Vt^2}{2} - (1+\lambda)kB - 16\kappa^2 t^2 \beta V + \{64\kappa^3 t^3 \alpha + (\lambda-1)k\}\tau_0, \tag{2.11}$$

$$P(v) = 64\kappa^3 t^3 \alpha v^{3/2} - 16\kappa^2 t^2 \beta v, \qquad \delta = \max\{0, P(a)\}. \tag{2.12}$$

Then

$$\prod_j |f_j(t)| \le \min\{1, e^{-S_\lambda}\}, \qquad \prod_{j\ne i} |f_j(t)| \le \min\{1, e^{-S_\lambda + m(t,a) + \delta}\}$$

for every index i.

Proof. The domain condition implies $r_j \ge 0$. Since $r_j - q_j \le k\epsilon_j$, the definition of φ_λ gives

$$\sum_j g(q_j) \ge \sum_j \varphi_\lambda(r_j) - \lambda k(B - \tau_0).$$

Apply (2.10) at $z_j = 4\kappa t\sqrt{v_j}$ and sum. This yields

$$\sum_j \varphi_\lambda(r_j) \ge 64\kappa^3 t^3 \alpha \tau_0 - 16\kappa^2 t^2 \beta V.$$

Together with $\sum_j q_j \ge H$, this bounds $\sum_j [q_j + g(q_j)]$ below by S_λ and proves the full-product inequality. For deletion, apply the same support directly to the remaining indices. The lost support term is $P(v_i)$, at most $\max\{P(0), P(a)\} = \delta$ by convexity. The linear sum loses at most $m(t,a)$. Replacing the remaining moment excess by $B - \tau_0$ can only decrease the exponent lower bound. This proves the second inequality. □

The next lemma reduces (2.10) to an endpoint inequality.

Lemma 2.3 (An endpoint criterion for a tangent lower bound)**.** *Let $\psi(z) = g(r(z))/z^2$, with its continuous value at zero, and $D(z) = z^4\psi''(z)$. Choose $0 < u < 2/3$ such that $r(u) \le q_\lambda$ and $D(u) \ge 0$. Set $\alpha = \psi'(u)$ and $\beta = u\psi'(u) - \psi(u)$. Then $\alpha, \beta \ge 0$. For $u \le M \le 1$, (2.10) follows from the single inequality*

$$\varphi_\lambda(r(M))/M^2 \ge \alpha M - \beta.$$

Proof. Direct differentiation gives

$$D = g''(r)\{Az^2(2-3z)\}^2 - 6rg'(r) + 6g(r),$$

$$D'(z) = \frac{4A^2z^3(2-3z)}{(1-2r)^3}\,B(z), \qquad B(z) = 1 - 6z + 2Az^2(3 - 5z + 3z^2).$$

The inequality $\max r < 1/2$ is equivalent to $A < 27/8$. On $[0, 2/3]$,

$$B'(z) = -6 + 6Az(2 - 5z + 4z^2) < 0,$$

because

$$\frac{8}{27} - z(2 - 5z + 4z^2) = (2/3 - z)(4z^2 - 7z/3 + 4/9) \ge 0.$$

The quadratic factor is positive. Also $B(0) = 1$ and $B(2/3) = -3 + 8A/9 < 0$. Thus D first increases and then decreases; $D(0) = 0$ and $D(2/3) = 6\{g(r) - rg'(r)\} < 0$. It has exactly one zero in $(0, 2/3)$. Hence ψ is initially convex and subsequently concave there. It is decreasing for $z \ge 2/3$.

On the capped branch,

$$\frac{\varphi_\lambda(r(z))}{z^2} = \lambda A(1-z) - \frac{c_\lambda}{z^2}, \qquad \left(\frac{\varphi_\lambda(r(z))}{z^2}\right)'' = -\frac{6c_\lambda}{z^4} \le 0.$$

Function and derivative match at each branch transition. Before $2/3$ the derivative therefore first increases and then decreases; afterwards it is nonpositive. The difference between $\varphi_\lambda(r(z))/z^2$ and the tangent at u decreases to zero, then increases, and finally decreases. The minimum on $[0, M]$ is consequently at u or M. Initial convexity and $\psi(0) = \psi'(0) = 0$ give $\alpha, \beta \ge 0$. □

The calculation uses tangent lower bounds with $\lambda \in \{3/4, 1, 3/2, 2, 3\}$. Arb checks the hypotheses and endpoint inequality of Lemma 2.3 for each stored set of coefficients. A bound is used only on its stated domain. In a parameter box, the exponent is affine in τ_0 and is minimized at an endpoint. The polynomial in (2.10) also determines the deletion term $P(v)$ in (2.12). When $\lambda < 1$, the evaluation bounds the nonnegative quantities $B + \tau_0$ and $B - \tau_0$ in $-k(B + \tau_0) - \lambda k(B - \tau_0)$ before expanding.

For the product comparisons below, write $\mathcal{F}_0$ and $\mathcal{D}_0$ for the minima of the elementary bounds (2.8) and all applicable bounds of Lemma 2.2. Each candidate is bounded over the entire parameter box before taking the minimum. The empty product has $\mathcal{F}_0 = \mathcal{D}_0 = 1$.

3 Real and imaginary zero-bias errors

The zero-bias transformation was introduced by Goldstein and Reinert [3]. For a centered variable of positive variance v, its zero-bias law is defined by

$$v\mathbb{E}g'(X^*) = \mathbb{E}[Xg(X)] \tag{3.1}$$

for absolutely continuous test functions for which the expectations exist. For summands, write $f_j^* = f_{X_j^*}$. The first Wasserstein distance between two laws is the infimum of $\mathbb{E}|U - V|$ over couplings with those laws. By the sharp zero-bias mean-distance bound of Goldstein [2, Lemma 1.1] and Tyurin [13, Theorem 3], the distance between X and X^* is at most $\mathbb{E}|X|^3/(2v)$. Applying this bound to e^{itx} gives

$$|f_X(t) - f_{X^*}(t)| \le |t|\frac{\mathbb{E}|X|^3}{2v}. \tag{3.2}$$

For bounds that also depend on the third-moment excess, let W have mean zero and variance one, and put $Y = |W|$, $\rho = \mathbb{E}Y^3$, and $e = \rho - 1$. Then $e \ge 0$ and

$$\mathbb{E}[(Y-1)^2(Y+1/2)] = e. \tag{3.3}$$

At $e = 0$, W has the Rademacher distribution: it takes the values -1 and 1 with probability $1/2$ each. Its characteristic function is $\cos x$, and that of its zero-bias law is $\operatorname{sinc} x$.

Lemma 3.1. *For every real x,*

$$|\Re f_W(x) - \cos x| \le e|x|^3/6, \qquad |\Re f_{W^*}(x) - \operatorname{sinc} x| \le e|x|/2, \tag{3.4}$$

where $\operatorname{sinc} x = \sin(x)/x$ with its continuous value at zero.

Proof. For an even function f with Lipschitz continuous second derivative, define

$$R_f(y) = f(y) - f(1) - \frac{f'(1)}{2}(y^2-1), \qquad y \ge 0.$$

Taylor's integral formula gives $R_f(y) = \int_0^{\max(1,y)} K_y(u) f'''(u)\,du$ with a nonnegative kernel K_y. If $y \le 1$, the kernel is $u(y-1)^2/2$ on $[0,y]$ and $(1-u)(u-y^2)/2$ on $[y,1]$. If $y \ge 1$, it is $u(y-1)^2/2$ on $[0,1]$ and $(y-u)^2/2$ on $[1,y]$. Its integral is $(y-1)^2(y+1/2)/6$. Apply this identity to $f(y) = \cos(xy)$, use $\|f'''\|_\infty \le |x|^3$, and take expectations in (3.3). The tangent's quadratic term has expectation zero because $\mathbb{E}Y^2 = 1$.

Define the square-biased law by $\mathbb{E}g(W^\square) = \mathbb{E}[W^2 g(W)]$ for bounded measurable g. The zero-bias representation is $W^* = UW^\square$, with U uniform on $[0,1]$ and independent of $W^\square$. For $z \ge 0$,

$$\mathbb{P}(|W^*| \le z) = \mathbb{E}[Y \min(z, Y)] \le \min(z, 1).$$

Thus $|W^*|$ stochastically dominates $\mathrm{Unif}[0,1]$. Their first Wasserstein distance is their mean difference $\rho/2 - 1/2 = e/2$. Apply the Lipschitz bound for $y \mapsto \cos(xy)$. □

Lemma 3.2. *For all real x,*

$$|\Im f_W(x)| \le |x|^3\sqrt{e(\rho + 5/3)}/6. \tag{3.5}$$

Proof. It suffices to take $x > 0$. Define $A(y) = x\operatorname{sinc}(xy)$. Centering gives

$$\Im f_W(x) = \mathbb{E}[W(A(Y) - A(1))].$$

Since $|\operatorname{sinc}'(z)| \le |z|/3$, we have $|A'(y)| \le x^3 y/3$ and hence

$$|A(y) - A(1)| \le (x^3/6)|y^2 - 1|.$$

Cauchy–Schwarz and (3.3) give

$$\mathbb{E}[Y|Y^2-1|] \le \sqrt{e\,\mathbb{E}\left[\frac{Y^2(Y+1)^2}{Y+1/2}\right]} \le \sqrt{e(\rho+5/3)}.$$

The final inequality follows from

$$\frac{y^2(y+1)^2}{y+1/2} = y^3 + \frac{3}{2}y^2 + \frac{y^2}{4(y+1/2)}, \qquad \frac{y^2}{y+1/2} \le \frac{2+4y^2}{9}.$$

The second inequality is equivalent to $(y-1)^2(4y+1) \ge 0$. □

4 A quadratic minorant for the real part

Lemma 4.1. *For $0 < x \le 2$, put*

$$F_x(y) = \cos(xy) - y\sin(xy)/x, \qquad A_x = \cos x + (1+x^2)\operatorname{sinc} x.$$

Then $F_x(y) \ge F_x(1) - A_x(y^2-1)/2$ for $y \ge 0$. Consequently

$$\Re(f_W(x) - f_{W^*}(x)) \ge \cos x - \operatorname{sinc} x. \tag{4.1}$$

Proof. Set $h_x(z) = \cos z + (1+x^2)\operatorname{sinc} z$ and $A_x = h_x(x)$. The tangent is $F_x(1) - A_x(y^2-1)/2$. Multiply the difference from this tangent by x^2 and put $z = xy$. Its derivative is $z[A_x - h_x(z)]$.

The function h_x is strictly decreasing on $(0,\pi)$. On $[\pi, 2\pi]$ it is at most one, and for $z \ge 2\pi$ it is at most $1 + 5/(2\pi) < 11/6$. Alternating Taylor lower bounds give

$$A_x \ge p_0(x^2), \qquad p_0(u) = 2 + u/3 - 7u^2/60 + 17u^3/2520 - u^4/5040.$$

For $0 \le u \le 4$, $p_0''(u) = (-98 + 17u - u^2)/420 < 0$, so $p_0(u) \ge \min(p_0(0), p_0(4)) = 194/105 > 11/6$. The tangent difference decreases up to $z = x$ and increases afterwards. Its minimum is zero. Taking expectations, and using $\mathbb{E}Y^2 = 1$, proves (4.1). □

For $0 \le x \le 2$, let $g = \operatorname{sinc} x - \cos x \ge 0$ and $c = x/2 + x^3/6$. Lemmas 3.1 and 4.1 give

$$-g \le \Re(f_W - f_{W^*}) \le -g + ce. \tag{4.2}$$

The alternating series gives the increasing upper bound

$$\frac{2g}{x} \le \alpha(x) := \begin{cases} 2x/3 - x^3/15 + x^5/420, & 0 \le x \le 2, \\ 2x/3, & x > 2. \end{cases} \tag{4.3}$$

The quotient at zero is defined by continuity. On $[0,2]$ the polynomial is at most $2x/3$. Put $q(x) = 1 + x^2/3$.

5 A quadratic bound in the third-moment excess

Let W have mean zero, variance one and $\rho = \mathbb{E}|W|^3 < \infty$. Write $e = \rho - 1$, $Y = |W|$, and $h(y) = (y-1)^2(y+1/2)$, so that $\mathbb{E}h(Y) = e$.

Lemma 5.1 (Quadratic imaginary-error estimate)**.** *For $0 < x \le 2$,*

$$|\Im(f_W(x) - f_{W^*}(x))|^2 \le e\left\{\frac{2}{3}[x\cos x - 2\sin x]^2 + \frac{x^2 e}{4}\right\}. \tag{5.1}$$

Proof. Put

$$A_x(y) = x\,\mathrm{sinc}(xy) + \frac{\cos(xy)}{x}, \qquad a_x = A_x'(1) = x\cos x - 2\sin x,$$

$$b_x = A_x''(1) = (2-x^2)\sin x - 3x\cos x, \qquad J_x(y) = \frac{y^2}{y+1/2}\left(\frac{A_x(y)-A_x(1)}{y-1}\right)^2.$$

Use continuous values at $y = 1$. Centering and the zero-bias identity give

$$\Im(f_W - f_{W^*}) = \mathbb{E}[W\{A_x(Y) - A_x(1)\}], \qquad |\Im(f_W - f_{W^*})|^2 \le e\mathbb{E}J_x(Y).$$

Moreover $J_x(1) = 2a_x^2/3$ and $J_x'(1) = 8a_x^2/9 + 2a_xb_x/3$. The scalar inequality established below is

$$J_x(y) \le J_x(1) + \frac{J_x'(1)}{2}(y^2-1) + \frac{x^2}{4}h(y) \qquad (y \ge 0). \tag{5.2}$$

Its expectation is at most $2a_x^2/3 + x^2e/4$, proving (5.1). All quantities are integrable: $|A_x'(y)| \le x(1+x^2/3)y$ gives $|A_x(y) - A_x(1)| \le (x/2 + x^3/6)|y^2-1|$, which bounds $J_x(y)$ by a constant times $y^2(y+1)^2/(y+1/2)$, which grows cubically. □

The polynomial verification of (5.2) is given in Appendix A.

5.1 Aggregation by a concave majorant

For $0 \le x \le 2$, use α and q defined above and put

$$p_1(x) = 1 + x^2/6 - x^4/40 + x^6/1008, \qquad b_0(x) = \tfrac{8}{3}p_1(x)^2.$$

The alternating series gives $2\,\mathrm{sinc}\,x - \cos x \le p_1(x)$ and $2(\mathrm{sinc}\,x - \cos x)/x \le \alpha(x)$. At $x = 0$, the quotient below denotes its continuous value $-\mathrm{i}\mathbb{E}W^3$. The real-part interval and Lemma 5.1 consequently imply

$$\left|\frac{2(f_W - f_{W^*})}{x}\right| \le \min\{1+e, \max(F_1(e), F_2(e))\}, \tag{5.3}$$

where

$$F_1(e) = \sqrt{\alpha^2 + b_0e + e^2}, \qquad F_2(e) = \sqrt{b_0e + (1+q^2)e^2}.$$

For the real part, use $\max\{\gamma^2, (qe-\gamma)^2\} \le \max\{\alpha^2, q^2e^2\}$ when $0 \le \gamma \le \alpha$.

Lemma 5.2 (The common-tangent envelope)**.** *Let $a, b \ge 0$, $0 \le u < v$, and $F_1(e) = \sqrt{a^2 + be + ue^2}$, $F_2(e) = \sqrt{be + ve^2}$ for $e \ge 0$. Their least concave majorant is F_2 when $a = 0$. If $a > 0$ and $b \le 2a\sqrt{v}$, it is $H(e) = a + \sqrt{v}\,e$. Otherwise put*

$$w = \sqrt{v-u}, \quad s = u/w^2, \quad r = b/(aw), \quad d_0 = \sqrt{r^2 - 4s}, \quad \eta = (d_0 - 2)/r,$$

$$c = \frac{r}{2\sqrt{d_0 + s - 1}}, \quad m = \frac{d_0 + 2s}{2\sqrt{d_0 + s - 1}}, \quad e_- = (a/w)\eta, \quad e_+ = (a/w)/\eta.$$

Then $d_0 > 2$, $0 < \eta < 1$, and

$$H(e) = \begin{cases} F_1(e), & 0 \le e \le e_-, \\ ac + wme, & e_- \le e \le e_+, \\ F_2(e), & e \ge e_+. \end{cases}$$

Proof. For $a > 0$, rescale by $z = we/a$. The squared branches become $1+rz+sz^2$ and $rz+(s+1)z^2$, crossing at $z = 1$. When $r \le 2\sqrt{s+1}$, the line $1 + \sqrt{s+1}z$ majorizes both by squaring. A concave majorant is at least one at zero and has limiting growth at least $\sqrt{s+1}z$; its chords to points tending to infinity force it above the same line.

In the remaining case both branches are concave. Direct algebra gives

$$(c + mz)^2 - (1 + rz + sz^2) = c^2(z - \eta)^2,$$

$$(c + mz)^2 - (rz + (s+1)z^2) = (c^2 - 1)(z - 1/\eta)^2, \qquad c^2 - 1 = \frac{(d_0 - 2)^2}{4(d_0 + s - 1)} > 0.$$

These identities prove global majorization and tangency. The piecewise function is concave with matching slopes. Every concave majorant must dominate the active branches and the chord between their contact points, so this one is least. For $a = 0$, $F_2 \ge F_1$ and F_2 is concave. □

Apply the lemma with $a = \alpha$, $b = b_0$, $u = 1$ and $v = 1 + q^2$, and put $\widehat{H}_x(e) = \min\{1 + e, H_x(e)\}$. A minimum of concave functions is concave. For a subarray with $v_i \le a$, Jensen's inequality with weights $v_i^{3/2}/\tau_0$ gives the prefactor

$$\sum_i v_i |f_i(t) - f_i^*(t)| \le \frac{t}{2} P_{\mathrm{quad}}(t; B, a, \tau_0), \qquad P_{\mathrm{quad}} = \tau_0 \widehat{H}_{t\sqrt{a}}\left(\frac{B - \tau_0}{\tau_0}\right). \tag{5.4}$$

Use $P_{\mathrm{quad}} = B$ if $\tau_0 = 0$ or $t\sqrt{a} > 2$. The coefficients increase on $[0, 2]$, and taking a least concave majorant preserves their pointwise order. Thus P_{quad} increases in t, a, B. All slopes of H_x exceed one and $\alpha(x) < 1$. Before the cap $1 + e$ is reached, $\widehat{H}_x - (1 + e)\widehat{H}'_x \le 0$; afterwards the ratio $\widehat{H}_x/(1 + e)$ is constant. Hence this ratio is nondecreasing, and the expression in (5.4) decreases in τ_0. These monotonicities give bounds on a parameter box by substitution of the appropriate endpoints.

For an alternative interval evaluation, the calculation also uses

$$\sum_j v_j |f_j(t) - f_j^*(t)| \le \frac{t}{2} P_{\mathrm{dir}}(t; B, a, \tau_0), \qquad t \ge 0, \tag{5.5}$$

where, with $x = t\sqrt{a}$,

$$P_{\mathrm{dir}}(t; B, a, \tau_0) = \min\left\{B, \sqrt{\alpha(x)^2\tau_0^2 + q(x)^2(B - \tau_0)(2B + 2\tau_0/3)}\right\}. \tag{5.6}$$

Indeed $P_{\mathrm{quad}} \le P_{\mathrm{dir}}$. For $0 \le x \le 2$, the function

$$R_x(e) = \sqrt{\alpha(x)^2 + \tfrac{8}{3}q(x)^2 e + 2q(x)^2 e^2}$$

majorizes both branches in (5.3), since $p_1 \le q$ and $q \ge 1$. It is concave in e: the quadratic's discriminant is $(64/9)q^4 - 8q^2\alpha^2 > 0$, because $\alpha^2 \le 4x^2/9 \le q^2/3$. The least concave majorant therefore satisfies $H_x \le R_x$. Applying the cap $1 + e$ proves the comparison. For $x > 2$, both prefactors equal B.

The bound P_{dir} increases with t, a, B and decreases with τ_0. For the latter assertion, expand the radicand as $2q^2B^2 - \frac{4}{3}q^2B\tau_0 + (\alpha^2 - 2q^2/3)\tau_0^2$; its nonconstant coefficients in τ_0 are nonpositive. Set $P_{\mathrm{dir}} = 0$ for the empty subset. Although it is weaker pointwise, its separate interval evaluation can give a smaller upper endpoint, so the calculation retains both enclosures.

5.2 Affine bounds in the third absolute moment

We also use affine bounds for

$$Z_W(x) = \frac{2(f_W(x) - f_{W^*}(x))}{x}, \qquad Z_W(0) = -\mathrm{i}\mathbb{E}W^3.$$

Put $\phi_2(z) = \int_0^1 (1-u)e^{uz}\,du$ and

$$h_x(w) = -2w^2(x + \mathrm{i}w)\phi_2(\mathrm{i}xw), \qquad h_0(w) = -\mathrm{i}w^3.$$

This is entire in x, w. The zero-bias representation and the moment identities give $\mathbb{E}h_x(W) = Z_W(x)$.

Suppose that, throughout a closed frequency cell and for every real w,

$$|h_x(w) - \mathrm{i}\beta w - \gamma(w^2 - 1)| \le c_0 + \delta + c_2|w|^2 + \Lambda|w|^3. \tag{5.7}$$

The moment identities annihilate the subtracted terms, so $|Z_W(x)| \le C + \Lambda\rho$, where $C = c_0 + c_2 + \delta$. For a common slope $\lambda \ge \Lambda > 0$, replace this by $(C + \Lambda - \lambda) + \lambda\rho$, since $\rho \ge 1$. Let $C_\lambda(X)$ be the maximum of the resulting intercepts over all frequency cells from zero to X. Summing the bound with slope λ over the summands gives

$$\sum_i v_i |f_i(t) - f_i^*(t)| \le \frac{t}{2}\{C_\lambda(X)\tau_0 + \lambda B\}, \qquad t\sqrt{a} \le X. \tag{5.8}$$

Each slope gives a bound for the sum; we may then take their minimum.

The table of modulus bounds covers $[0, 2]$ in 128 closed cells of width $1/64$ at slopes $5/4, 3/2, 2, 3, 4$. Its certificates establish (5.7) on a compact rectangle by proving both $P \ge 0$ and $P^2 - |h_x - \mathrm{i}\beta w - \gamma(w^2-1)|^2 \ge 0$, using outward interval and Taylor estimates. All real w are covered by an analytic tail. Here P denotes the right-hand side of (5.7). For example, if $\Lambda > 1$ and $0 \le x \le X$, $|\phi_2(\mathrm{i}xy)| \le 1/2$ gives

$$|h_x(y)| \le y^3 + X^2 y/2.$$

For $y \ge R \ge 1$, a lower tail polynomial for the difference in (5.7) is

$$(\Lambda - 1)y^3 + (c_2 - |\gamma|)y^2 - (X^2/2 + |\beta|)y + c_0 + \delta + |\gamma|.$$

Nonnegative exact coefficients after $y = R + z$ prove the entire tail. For a positive frequency cell $[x_0, x_1]$ and $0 < \Lambda \le 1$, use

$$G = \max\{|\gamma - 2/x_0|, |\gamma - 2/x_1|\}, \quad B_0 = |\beta + 2|,$$

$$\Lambda y^3 + (c_2 - G)y^2 - (4/x_0^2 + B_0)y + c_0 + \delta + G - 2/x_0$$

instead. Define the equivalent trigonometric target

$$g_x(y) = \frac{2}{x}\left\{e^{\mathrm{i}xy} + \frac{\mathrm{i}y}{x}(e^{\mathrm{i}xy} - 1)\right\}, \qquad h_x(y) = g_x(y) - 2\mathrm{i}y - \frac{2}{x}(1 - y^2).$$

The bound $|g_x(y)| \le 2/x_0 + 4y/x_0^2$ proves the claimed tail polynomial.

A directional affine table, retained as an additional minimum, certifies real projections of the same target in 48 exact rational directions. The stored certificates use 25 directions in the closed upper half-plane, including both real-axis directions. Reflecting the other 23 directions gives the

full set of 48: the identity $h_x(-w) = \overline{h_x(w)}$ supplies the reflected projection bounds by replacing w with $-w$ in their polynomial majorants. Their convex hull contains the disk of radius $\eta = 997/1000$. For each direction ζ, a real-line polynomial majorizes $\Re(\zeta h_x(w))$; its expectation uses only $\mathbb{E}W = 0$, $\mathbb{E}W^2 = 1$ and $\mathbb{E}|W|^3 = \rho$. Maximizing the intercepts over all directions yields the bound in (5.8) divided by η. The polygon inclusion is checked by exact supporting half-planes. This factor belongs only to the directional table. For box evaluation, an affine intercept of either sign uses the endpoint of τ_0 that maximizes it.

In the following product estimates, $P(t; B, a, \tau_0)$ denotes the minimum of P_{dir}, P_{quad} and the applicable bounds (5.8). It satisfies

$$\sum_i v_i |f_i(t) - f_i^*(t)| \le \frac{t}{2} P(t; B, a, \tau_0).$$

Only estimates valid on the complete argument interval may enter the minimum. For the empty subset, $P = 0$.

5.3 A characteristic-function bound using moment excess

Squaring the real-part estimate in Lemma 3.1 and adding the square of the bound in (3.5) gives another single-summand bound. If $\tau_j = v^{3/2}$ and $C = |\cos(t\sqrt{v})|$, then

$$|f_X(t)|^2 \le C^2 + \frac{C(b-\tau_j)|t|^3}{3} + \frac{t^6(b-\tau_j)(2b+2\tau_j/3)}{36}. \tag{5.9}$$

This holds for every real t. We use the smaller of its square root and (2.4), evaluated at $|t|$. The displayed terms increase with $C, b, |t|$ and decrease with τ_j when $b \ge \tau_j$. On a parameter interval, first bound C over the entire cosine argument interval, then apply these monotonicities.

6 A summand of maximal variance

The independent-sum replacement rule of Goldstein and Reinert [3, Lemma 2.1(5)], with total variance one, gives

$$f_{S^*}(t) = \sum_i v_i f_{X_i^*}(t) \prod_{j \ne i} f_{X_j}(t). \tag{6.1}$$

For the remaining summands, use the product bound

$$R(t) = \min\{e^{-\psi(t;w,B_R+\tau_R)}, \mathcal{F}_0(t; w, B_R, a_R, \tau_R)\}.$$

If the remainder is empty, set $R(t) = D_R(t) = 1$. Let $A(t)$ be the minimum of the single-summand bounds (2.4) and the square root of (5.9) for X_1. Let $D_R(t)$ be the bound $\mathcal{D}_0$ for the remaining summands, and $D_S(t)$ the corresponding bound for the whole sum. We also evaluate these bounds with Q omitted and take the smaller interval upper bounds. The defining inequality for P gives

$$\begin{aligned} |f_S(t) - f_{S^*}(t)| \le \frac{t}{2}\{&P(t; b, d, d^{3/2})R(t) \\ &+ P(t; B_R, a_R, \tau_R)A(t)D_R(t)\}. \end{aligned} \tag{6.2}$$

The whole-sum estimate is $(t/2)P(t; L, d, \tau)D_S(t)$, and the classical estimate is $tL/2$. Thus $|f_S(t) - f_{S^*}(t)| \le (tL/2)\mathcal{B}(t)$, where

$$\mathcal{B}(t) = \min\left\{1, \frac{P(t; L, d, \tau)D_S(t)}{L}, \frac{P(t; b, d, d^{3/2})R(t) + P(t; B_R, a_R, \tau_R)A(t)D_R(t)}{L}\right\}. \tag{6.3}$$

This formula also defines the endpoint value $\mathcal{B}(0)$.

For the characteristic function itself, use

$$F_S(t) = \min\{e^{-\psi(t;1,L+\tau)},\ A(t)R(t),\ \mathcal{F}_0(t;1,L,d,\tau)\}. \tag{6.4}$$

For another bound on the last term of (6.3), put $\lambda = b/L$ and

$$U_1 = \frac{P(t;b,d,d^{3/2})R(t)}{b}, \qquad U_R = \frac{P(t;B_R,a_R,\tau_R)A(t)D_R(t)}{B_R}.$$

If $B_R = 0$, set $U_R = 0$. Both quantities lie in $[0,1]$, and the last term in (6.3) equals $\lambda U_1 + (1-\lambda)U_R$. Suppose a box gives $\lambda \in [\lambda_-, \lambda_+] \subseteq [0,1]$ and $U_i \le \overline{U}_i$. Then

$$\lambda U_1 + (1-\lambda)U_R \le \max_{\lambda \in \{\lambda_-, \lambda_+\}} \{\lambda \overline{U}_1 + (1-\lambda)\overline{U}_R\}. \tag{6.5}$$

The right side is the maximum of an affine function on an interval. We take the smaller of this bound and a direct interval evaluation of the numerator in (6.3), divided by a lower bound for L. When a denominator in the separate bounds for U_i has lower endpoint zero, the bound $U_i \le 1$ is used.

Following the comparison in Tyurin [13, Lemma 3], write $D(t) = f_S(t) - e^{-t^2/2}$. The zero-bias identity gives $f_S'(t) = -t f_{S^*}(t)$, and hence

$$D'(t) + tD(t) = t(f_S(t) - f_{S^*}(t)), \qquad D(0) = 0.$$

Multiplication by $e^{t^2/2}$ and integration yield

$$D(t) = e^{-t^2/2} \int_0^t u(f_S(u) - f_{S^*}(u)) e^{u^2/2}\, du.$$

Taking absolute values gives

$$\frac{|f_S(t) - e^{-t^2/2}|}{L} \le R_\Delta(t) := e^{-t^2/2} \int_0^t \frac{u^2}{2} \mathcal{B}(u) e^{u^2/2}\, du. \tag{6.6}$$

7 Comparison with a cosine product

We also compare f_S with the characteristic function of a sum of symmetric two-point variables having the same variances. This intermediate approximation is studied by Mattner and Shevtsova [6, Theorem 1.2 and Example 1.4]. Here a differential identity bounds the error, and the real displacement from the Gaussian is retained in the smoothing integral. Set $g(t) = e^{-t^2/2}$ and write

$$C_v(t) = \prod_i \cos(t\sqrt{v_i}), \quad k_v(t) = \sum_i \sqrt{v_i} \tan(t\sqrt{v_i}), \quad r_i(t) = f_i'(t) + \sqrt{v_i} \tan(t\sqrt{v_i}) f_i(t).$$

For $t\sqrt{d} < \pi/2$, $C_v(t) > 0$ and

$$(f_S - C_v)' + k_v(f_S - C_v) = \sum_i r_i \prod_{j \ne i} f_j, \qquad f_S(0) - C_v(0) = 0. \tag{7.1}$$

The identity follows from the product rule.

7.1 The forcing term

For a standardized variable and $0 < x < \pi/2$, define

$$T_x(w) = \frac{2}{x^2}\{(\mathrm{i}w + \tan x)e^{\mathrm{i}xw} - \mathrm{i}w\sec x - x(1-w^2)\}, \qquad T_0(w) = \mathrm{i}(w - w^3). \tag{7.2}$$

Then

$$\mathbb{E}T_x(W) = 2\{f_W'(x) + \tan(x) f_W(x)\}/x^2, \qquad r_i(t) = \tfrac{1}{2}t^2 v_i^{3/2}\mathbb{E}T_{t\sqrt{v_i}}(W_i).$$

The elementary estimate, including its continuous value at zero, is

$$\begin{aligned} |\mathbb{E}T_x(W)| &\le 2p(x)\sqrt{(\rho-1)(\rho+5/3)} \quad (0 \le x < \pi/2), \\ p(x) &= \frac{x\tan x + 1 - \sec x}{x^2}, \quad p(0) = \tfrac{1}{2}. \end{aligned} \tag{7.3}$$

To prove it, write $q_0(u) = f_W''(u) + f_W(u) = \mathbb{E}[(1-W^2)(e^{iuW}-1)]$. The moment-excess Cauchy inequality gives $|q_0(u)| \le u\sqrt{(\rho-1)(\rho+5/3)}$. The Wronskian identity

$$f_W'(x)\cos x + f_W(x)\sin x = \int_0^x q_0(u)\cos u\, du$$

and positivity of $\cos u$ prove (7.3). Also $p(x) = \int_0^1 r\cos(rx)/\cos x\, dr$ is positive and increasing below $\pi/2$.

The scalar inequality in Appendix B gives

$$|\mathbb{E}T_x(W)|^2 \le (\rho-1)\{q_{\mathrm{ref}}(x) + \tfrac{9}{8}(\rho-1)\}, \qquad q_{\mathrm{ref}}(x) = \frac{8}{3}\frac{\operatorname{sinc}^4 x}{\cos^2 x}, \quad 0 \le x \le 1. \tag{7.4}$$

Let $D_{\mathrm{del}}(t)$ bound every deleted product. With $w_i = v_i^{3/2}$ and $e_i = b_i - w_i$, Cauchy–Schwarz gives

$$\sum_i w_i|\mathbb{E}T_{t\sqrt{v_i}}| \le \sqrt{(L-\tau)\{q_{\mathrm{ref}}(t\sqrt{d})\tau + \tfrac{9}{8}(L-\tau)\}}$$

when $t\sqrt{d} \le 1$. Multiplication by $t^2 D_{\mathrm{del}}(t)/2$ yields a forcing bound for (7.1). The elementary alternative is

$$t^2 D_{\mathrm{del}}(t) p(t\sqrt{d})\sqrt{(L-\tau)(L+5\tau/3)}.$$

We can also retain the individual values of q_{ref} in this estimate. Set $a = d^{3/2}$, $f_t(w) = wq_{\mathrm{ref}}(tw^{1/3})$, $k = \lfloor \tau/a \rfloor$, $r = \tau - ka$, and $F_t(a,\tau) = kf_t(a) + f_t(r)$. The function f_t is increasing and convex: indeed

$$q_{\mathrm{ref}}(x) = \frac{8}{3}\left(\frac{\sec x - \cos x}{x^2}\right)^2$$

has nonnegative even Taylor coefficients. Positivity follows from $\tan' = 1 + \tan^2$ and $\sec' = \sec\tan$; additionally $\sec'' = \sec + 2\sec\tan^2$ implies that the coefficient of x^{2n} in $\sec x$ is at least $1/(2n)!$. Hence $f_t(w) = \sum_{n\ge 0} c_n t^{2n} w^{1+2n/3}$ with $c_n \ge 0$. Pushing two weights in $(0,a)$ to their available endpoints cannot decrease their sum under a convex function. Iteration gives

$$\sum_i w_i q_{\mathrm{ref}}(tw_i^{1/3}) \le F_t(a,\tau).$$

This gives the forcing bound

$$\frac{t^2 D_{\mathrm{del}}(t)}{2}\sqrt{(L-\tau)\{F_t(a,\tau) + \tfrac{9}{8}(L-\tau)\}}. \tag{7.5}$$

The maximizing weights need not satisfy the total-variance identity; maximizing over this larger set preserves an upper bound. We use the minimum of the applicable forcing bounds. For $1 < t\sqrt{d} < \pi/2$, the same differential inequality uses the elementary bound and the preceding error bound as its initial value.

7.2 Using the third moment of the largest summand

The distinguished third moment b can improve the forcing bound. Write

$$w = d^{3/2}, \quad e = b - w, \quad R = \tau - w, \quad e_R = L - b - R, \quad E = L - \tau = e + e_R, \quad \delta = Lw - b\tau.$$

For an actual array, w, e, R, e_R are nonnegative. Suppose a common scalar estimate has the form $|\mathbb{E}T_x(W_i)|^2 \le (\rho_i - 1)\{Q + \lambda(\rho_i - 1)\}$, where $Q > 0$ and $\lambda \ge 0$. Applying Cauchy–Schwarz only to the remaining summands gives the moment factor

$$S_Q = \sqrt{e(Qw + \lambda e)} + \sqrt{e_R(QR + \lambda e_R)}. \tag{7.6}$$

This bounds the forcing term in (7.1) by

$$\left| \sum_i r_i(t) \prod_{j \ne i} f_j(t) \right| \le \frac{t^2 D_{\mathrm{del}}(t)}{2} S_Q.$$

The same multiplier applies to the convex weight factor S_P below. Put $A = Qw + \lambda e$, $B = QR + \lambda e_R$, and $G^2 = E(Q\tau + \lambda E)$. The identities

$$G^2 - S_Q^2 = (\sqrt{eB} - \sqrt{e_R A})^2, \qquad eB - e_R A = -Q\delta$$

and $(\sqrt{eB} + \sqrt{e_R A})^2 \le G^2$ imply, when $\delta > 0$,

$$S_Q \le G\sqrt{1 - \left\{ \frac{\delta}{E(\tau + (\lambda/Q)E)} \right\}^2}. \tag{7.7}$$

Zero forcing is treated directly, without division.

For the convex weight bound, set $q = q_{\mathrm{ref}}$, $\lambda = 9/8$, $q_j = q(tw^{1/3})$, and $F_R = F_t(w, R)$. The corresponding factor is

$$S_P = \sqrt{e(q_j w + \lambda e)} + \sqrt{e_R(F_R + \lambda e_R)}. \tag{7.8}$$

Since the largest weight w is attained, $F_t(w, \tau) = wq_j + F_R$. Apply the same identity with $A = q_j w + \lambda e$ and $B = F_R + \lambda e_R$. Monotonicity of q gives $F_R \le q_j R$, hence $eB - e_R A \le -q_j \delta$ when $\delta \ge 0$. As $q_j \ge 8/3$, this proves

$$S_P \le \sqrt{E\{F_t(w, \tau) + \tfrac{9}{8}E\}} \sqrt{1 - \left\{ \frac{\delta}{E(\tau + 27E/64)} \right\}^2} \qquad (\delta \ge 0). \tag{7.9}$$

The same last factor applies to (7.7) whenever $\lambda/Q \le 27/64$. The elementary estimate has $\lambda/Q = 3/8$; the scalar-table estimates used with this factor satisfy the displayed inequality by exact comparison of their coefficients.

We next give the interval bounds used to maximize (7.6) and (7.8). For the convex weight bounds, assume $t\sqrt{d_+} \le 1$. The generic split bound applies wherever its scalar inequality holds. For a parameter box, choose $w_- \le d_-^{3/2}$ and $w_+ \ge d_+^{3/2}$, and put $M = L_+$ and $a = \tau_-$. Suppose

$$a - w_+ \ge \frac{15}{23}(M - b_-). \tag{7.10}$$

Both split factors increase with L. With w, b fixed, increase L to M and then decrease τ to a. To justify the second step, put $H = M - b$ and differentiate the squared remainder term with respect to R:

$$\frac{d}{dR}\{(H-R)[QR + \lambda(H-R)]\} = Q(e_R - R) - 2\lambda e_R \le 0 \quad (R \ge H/2).$$

For the weight bound, $F_t(w,R) \ge 8R/3$ and $0 \le \partial_R F_t(w,R) < 29/4$ on each interval between consecutive multiples of w. Thus

$$\frac{d}{dR}\{(H-R)[F_t(w,R) + \tfrac{9}{8}(H-R)]\} \le 5e_R - \frac{8}{3}R \le 0 \quad (R \ge 15H/23).$$

The function $F_t(w,R)$ is continuous at the joins. Condition (7.10) therefore proves

$$S_Q(L,\tau,w,b) \le S_Q(M,a,w,b), \qquad S_P(L,\tau,w,b) \le S_P(M,a,w,b).$$

All moments in these expressions remain nonnegative along the comparison; the endpoints are used only to bound the algebraic factors. Set

$$E_* = M - a, \qquad \delta_* = Mw_- - b_+a, \qquad D_* = E_*(a + 27E_*/64).$$

For S_P, and for S_Q with $\lambda/Q \le 27/64$, suppose $0 < \delta_* \le D_*$. Any rational $\eta \in (0,1]$ satisfying $\eta^2 + (\delta_*/D_*)^2 \ge 1$ gives

$$S_Q(L,\tau,w,b) \le \eta\sqrt{E_*(Qa + \lambda E_*)},$$
$$S_P(L,\tau,w,b) \le \eta\sqrt{E_*\{F_t(w,a) + 9E_*/8\}}.$$

A larger weight cap can replace w in $F_t(w,a)$ wherever that function is defined. Under the same coefficient-ratio restriction for S_Q, a direct bound on the whole box is also available: use $\delta_* = L_-w_- - b_+\tau_+$ and $D_* = (L_+ - \tau_-)\{\tau_+ + 27(L_+ - \tau_-)/64\}$ in the same test, without requiring (7.10).

There is a less restrictive endpoint condition when the remaining weight is smaller than w throughout the box. Write $R_- = a - w_+$, $R_+ = \tau_+ - w_-$, and take $0 < W_0 \le d_+^{3/2}$. If

$$0 < R_- \le R_+ < w_-, \qquad R_+ \le (19/25)^3 W_0, \qquad R_- \ge \frac{23}{55}(M - b_-), \tag{7.11}$$

then the same endpoint comparison holds for S_P. Indeed $F_t(w,R) = f_t(R)$ throughout the comparison, and $tR^{1/3} \le 19/25$. The derivative bound $f_t'(R) < 25/6$ gives

$$\frac{d}{dR}\{e_R[f_t(R) + \tfrac{9}{8}e_R]\} \le \frac{23}{12}e_R - \frac{8}{3}R \le 0.$$

This condition applies to the weight bound alone.

Both derivative constants follow from exact rational bounds. The nonnegative even coefficients of q make $g(x) = q(x) + xq'(x)/3$ increasing, and direct differentiation gives

$$g(x) = \frac{8\sin^3 x\{2x(1+\cos^2 x) - \sin x\cos x\}}{9x^4\cos^3 x}.$$

Let s_- and s_+ be the sine polynomials through degrees seven and nine, and let c_- and c_+ be the cosine polynomials through degrees ten and twelve. Alternating-series bounds give

$$g(x) \le \frac{8s_+(x)^3\{2x(1+c_+(x)^2) - s_-(x)c_-(x)\}}{9x^4c_-(x)^3}.$$

At $x = 10001/10000$ this rational expression is less than $29/4$; at $x = 19/25$ it is less than $25/6$. Monotonicity gives the bounds on the full intervals, including $g(0) = 8/3$.

7.3 A quantitative bound for the convex weight maximum

The weight maximum and the Cauchy–Schwarz maximum need not occur at the same distinguished weight. We bound their combined effect. Assume one of the endpoint conditions above applies to S_P, and put $E = M - a > 0$ and $\lambda = 9/8$. Let $W \geq d_+^{3/2}$ be the weight cap used in the interval evaluation. Choose c, W_0 such that

$$b_+ a/M < c < W_0 \leq W,$$

and suppose $\lfloor a/u \rfloor = k \geq 1$ on (c, W_0). Choose $d_0 > \gamma \geq 0$ with

$$u^{2/3} \geq d_0 > \gamma \geq (a - ku)^{2/3} \qquad (c \leq u \leq W_0).$$

The coefficient expansion is

$$q(x) = \sum_{n \geq 0} q_n x^{2n}, \qquad q_0 = \frac{8}{3}, \quad q_1 = \frac{8}{9}, \quad q_2 = \frac{8}{15}, \quad q_3 = \frac{736}{2835}, \qquad q_n \geq 0.$$

Consequently, for any finite $N \geq 1$,

$$\frac{d}{du} F_t(u, a) = k\{f_t'(u) - f_t'(a - ku)\} \geq k \sum_{n=1}^{N} q_n (1 + 2n/3)(d_0^n - \gamma^n) t^{2n}.$$

Integrate from c to W_0 and use monotonicity in the cap beyond W_0. The endpoint joins are covered by continuity. Take a frequency interval $[l, r]$ satisfying

$$0 \leq l \leq r, \qquad r\sqrt{d_+} \leq 1, \qquad rW^{1/3} \leq 10001/10000.$$

These conditions also keep the relevant cosine factors positive. Choose $\bar{q} \geq q(rW^{1/3})$ and set

$$\begin{aligned} A &= \min\{(29/4)a + \lambda E,\ a\bar{q} + \lambda E\}, \\ P_N(y) &= \frac{k(W_0 - c)}{A} \sum_{n=1}^{N} q_n (1 + 2n/3)(d_0^n - \gamma^n) y^n, \\ H &= \left\{ \frac{Mc - b_+ a}{E(a + 27E/64)} \right\}^2. \end{aligned}$$

The derivative bound $g < 29/4$ and the bound $\bar{q}$ each give an upper bound for $F_t(W, a)/a$, which justifies the minimum defining A. Require also $H \leq 1$. Since $E\{F_t(W, a) + \lambda E\} \leq EA$, the integrated inequality gives, for $w \leq c$,

$$S_P(M, a, w, b)^2 \leq \{1 - P_N(t^2)\}\, E\{F_t(W, a) + \lambda E\}.$$

For $w \geq c$, the defect is at least $Mc - b_+ a > 0$, and (7.9) gives the same bound with H in place of $P_N(t^2)$. Thus throughout the box and frequency cell,

$$S_P(L, \tau, w, b) \leq \left(1 - \frac{1}{2} \min\{P_N(l^2), H\}\right) \sqrt{E\{F_t(W, a) + \lambda E\}}. \tag{7.12}$$

Here we used the nonnegative coefficients of P_N and $\sqrt{1 - x} \leq 1 - x/2$ for $0 \leq x \leq 1$. Several valid choices of c, W_0, d_0, γ may be used: the largest of their lower bounds for the loss remains valid.

7.4 Bounds for the cosine product

Lemma 7.1. *Let $M = \sqrt{d} < 1$ and $m = (\tau - M^3)/(1 - M^2)$. For every increasing convex function h on $[0, M]$,*

$$\sum_i v_i h(\sqrt{v_i}) \geq M^2 h(M) + (1 - M^2) h(m).$$

In particular, below the cosine pole,

$$\begin{aligned} C_v(t) &\leq C_*(t) := \cos(tM) \cos(tm)^{(1-M^2)/m^2}, \\ k_v(t) &\geq K_*(t) := M \tan(tM) + (1 - M^2) \tan(tm)/m, \\ C_v(r)/C_v(l) &\leq C_*(r)/C_*(l) \qquad (0 \leq l \leq r). \end{aligned}$$

The formulas have Gaussian limits at $m = 0$; $d = 1$ is the exact singleton.

Proof. Remove one attained largest-variance coordinate and apply weighted Jensen to the rest, whose total variance is $1 - M^2$ and weighted mean of $\sqrt{v_i}$ is m. Necessary moment inequalities give $0 \leq m \leq M$. The three applications use respectively

$$-\log\cos(tx)/x^2, \quad \tan(tx)/x, \quad [\log\cos(lx) - \log\cos(rx)]/x^2.$$

Their continuous values at zero exist. Nonnegative Taylor coefficients below the pole make each increasing and convex. Applying Jensen to the frequency difference proves the ratio inequality directly. □

For box evaluation the lower corner must remain feasible. Put

$$D_0 = \max\{d_{lo}, \tau_{lo}^2\}, \quad M_0 = \sqrt{D_0}, \quad \tau_0 = \max\{\tau_{lo}, M_0^3\}, \quad m_0 = (\tau_0 - M_0^3)/(1 - D_0)$$

when $D_0 < 1$. If $F_h(M, \tau) = M^2 h(M) + (1 - M^2) h((\tau - M^3)/(1 - M^2))$, differentiation on the feasible region gives

$$\partial_M F_h = 2M\{h(M) - h(m) - (M - m)h'(m)\} + M^2\{h'(M) - h'(m)\} \geq 0, \qquad \partial_\tau F_h = h'(m) \geq 0.$$

Starting from (M_0, τ_0), increase M along $\tau(M) = \max(\tau_0, M^3)$ and then increase τ at the final M. On the boundary $\tau = M^3$, the expression is $M^2 h(M) + (1 - M^2) h(0)$, also increasing. Thus the lower corner supplies an upper bound for C_v, a lower bound for k_v, and an upper bound for $C_v(r)/C_v(l)$ throughout the parameter box.

To bound $C_v - g$, let $J(z) = -\log\cos z - z^2/2$ and

$$A_*(t) = D_0 \frac{J(tM_0)}{M_0^2} + (1 - D_0) \frac{J(tm_0)}{m_0^2},$$

with zero correction when a denominator variable is zero. On a cell $[l, r]$,

$$g(l) \operatorname{expm1}\left[-\frac{\tau_{hi}}{d_{hi}^{3/2}} J(r\sqrt{d_{hi}})\right] \leq C_v(t) - g(t) \leq g(r) \operatorname{expm1}[-A_*(l)] \leq 0. \tag{7.13}$$

The lower bound follows from the chord upper bound for $-\log\cos(tx)/x^2$ on $[0, \sqrt{d}]$ with weights v_i. The upper bound follows from Lemma 7.1. Positive correction series are increasing in their arguments; factoring out g proves (7.13) without assuming that the offset itself is monotone in frequency.

If A_j bounds the modulus of the right-hand side of (7.1) on $[l, r]$, P_j bounds $|f_S(l) - C_v(l)|$, and k_j is a lower bound for k_v on the cell, put $h = r - l$ and $\Psi_j = (1 - e^{-k_j h})/k_j$, with value h at zero. Comparison with $z' = -k_j z + A_j$ gives

$$E_j = \max\{P_j, e^{-k_j h} P_j + A_j \Psi_j\}, \qquad P_{j+1} \le Q_j P_j + A_j \Psi_j, \tag{7.14}$$

where Q_j bounds $C_v(r)/C_v(l)$. Here E_j bounds $|f_S - C_v|$ throughout the cell, and P_{j+1} bounds it at the right endpoint. The calculation takes the minimum of the applicable error bounds and uses (6.6) wherever the cosine comparison does not apply.

The uniform factor from Section 7.2 can be applied to a stored radius when it bounds every forcing candidate used to construct that radius. This follows by induction from zero through the nonnegative, positively homogeneous operations in (7.14). The factor in (7.12) varies with the frequency cell and bounds only the convex weight candidate. We apply it to that forcing candidate and evaluate the recurrence with the new source minima. In both cases the signed center $C_v - g$ is given by (7.13); the contraction multiplies only the radius of the error about that center.

8 Smoothing and numerical integration

For $0 < u < 1$, put

$$K(u) = \frac{1 - u}{2} + \frac{\mathrm{i}}{2}\left((1 - u)\cot(\pi u) + \frac{1}{\pi}\right).$$

The smoothing inequality of Prawitz [9], in the form stated by Shevtsova [12, Lemma 4.1] and Tyurin [13, eq. (46)], gives for any $T > 0$ and $0 < s < 1$

$$\frac{\Delta}{L} \le 2\int_0^s |K(u)| R_\Delta(Tu)\, du + \frac{2\int_s^1 |K(u)| F_S(Tu)\, du + G(T, s)}{L}, \tag{8.1}$$

where

$$G(T, s) = 2\int_0^s \left|K(u) - \frac{\mathrm{i}}{2\pi u}\right| e^{-T^2u^2/2}\, du + \frac{E_1(T^2 s^2/2)}{2\pi}, \qquad E_1(z) = \int_z^\infty \frac{e^{-v}}{v}\, dv. \tag{8.2}$$

8.1 Integration over a frequency interval

On a frequency interval $[a, b]$, let $\mathcal{B}_j$ be an upper bound for $\mathcal{B}$, and write

$$I_a = \int_0^a u^2 \mathcal{B}(u) e^{u^2/2}/2\, du.$$

Let A_j be an upper bound for $2\pi u|K(u)|$ on $u \in [a/T, b/T]$. Under the change of variable $t = Tu$, the first term of (8.1), restricted to $[a, b]$, is at most

$$\frac{A_j}{\pi}\left\{I_a \frac{E_1(a^2/2) - E_1(b^2/2)}{2} + \mathcal{B}_j\left(\frac{b^3 - a^3}{18} - \frac{a^3}{6}\log\frac{b}{a}\right)\right\}. \tag{8.3}$$

To see this, split the integral defining $R_\Delta(t)$ at a. The part up to a contributes $\int_a^b I_a e^{-t^2/2}/t\, dt$, which equals the first term in braces. For the remaining part, use $e^{-(t^2-u^2)/2} \le 1$ and interchange the nonnegative integrals:

$$\int_a^b \frac{1}{t}\int_a^t \frac{u^2}{2}\, du\, dt = \frac{b^3 - a^3}{18} - \frac{a^3}{6}\log\frac{b}{a}.$$

At $a = 0$, $I_a = 0$ and the expression in braces in (8.3) is $\mathcal{B}_j b^3/18$. In the calculation, I_a is replaced by an upper bound accumulated from the preceding intervals. For $a > 0$, we also integrate the bound $|f_S - e^{-t^2/2}| \le F_S + e^{-t^2/2}$ over $[a, b]$ and use the smaller of the two upper bounds for that integral.

8.2 The Gaussian correction

Direct subtraction of the singular part of K gives

$$2\left|K(u) - \frac{\mathrm{i}}{2\pi u}\right| = (1-u)\sqrt{1+h(u)^2}, \qquad h(u) = \frac{1}{\pi u} - \cot(\pi u). \tag{8.4}$$

Set $h(0) = 0$ by continuity. The partial-fraction identity [7, eq. 4.22.3] is

$$h(u) = \frac{2u}{\pi}\sum_{n=1}^{\infty}\frac{1}{n^2-u^2}.$$

For $0 \le u \le s < 1$ and any integer $M \ge 1$, define

$$c_M(s) = \frac{\pi^2/6 - \sum_{n=1}^{M} n^{-2}}{1 - s^2/(M+1)^2}, \qquad h_M(u) = \frac{2u}{\pi}\left(\sum_{n=1}^{M}\frac{1}{n^2-u^2} + c_M(s)\right).$$

For $n \ge M+1$ and $0 \le u \le s$, $1/(n^2-u^2) \le n^{-2}/(1-s^2/(M+1)^2)$. Summing gives $0 \le h(u) \le h_M(u)$. In the computation, $c_M(s)$ is replaced by an upper dyadic endpoint. Since $s < 1$, every denominator in the finite sum is positive on $[0, s]$. Hence

$$G(T,s) \le \int_0^s (1-u)\sqrt{1+h_M(u)^2}\,e^{-T^2u^2/2}\,du + \frac{E_1(T^2s^2/2)}{2\pi}. \tag{8.5}$$

The calculation uses $M = 8$ and Arb to enclose the integral in (8.5); the integration routine is described in Section 10.3. We also evaluate the following bound and use the smaller upper endpoint:

$$G(T,s) \le \int_0^s (1-u+\pi^2u^2/18)e^{-T^2u^2/2}\,du + \frac{E_1(T^2s^2/2)}{2\pi}, \tag{8.6}$$

also supplied by Shevtsova [12, Lemma 4.1]. Its integral is expressible using the error function and elementary functions.

9 Smoothing at a fixed threshold

Use $k_T(t) = K(t/T)/T$ and $g(t) = e^{-t^2/2}$. With the convention $f_S(t) = \mathbb{E}e^{\mathrm{i}tS}$, the signed inequality of Prawitz [9] gives

$$\mathbb{P}\{S \le x\} \le \tfrac{1}{2} + 2\Re\int_0^T k_T(t)e^{-\mathrm{i}xt}f_S(t)\,dt. \tag{9.1}$$

We use the formulation in Pinelis [8, eqs. (2.2)–(2.6)], which applies to the distribution function at atoms as well. Its filter satisfies $\mathrm{i}M(u)/(2\pi u) = K(u)$ for $u > 0$. The integral is the limit of its real part at zero.

Define

$$\Gamma_{T,a}(x) = \tfrac{1}{2} + 2\Re\int_0^a k_T(t)e^{-\mathrm{i}xt}g(t)\,dt - \Phi(x).$$

On the grid $t_j = Tsj/N$, choose bounds

$$C_j \geq \frac{2}{L}\int_{t_j}^{t_{j+1}} |k_T(t)|\,|f_S(t) - g(t)|\,dt$$

uniformly on the parameter box. Such bounds follow from (6.6) and $|f_S - g| \leq F_S + g$. Let A_j bound $2\int_{t_j}^{t_{j+1}} |k_T f_S|$ and let H bound this integral over $[Ts, T]$. Splitting (9.1) at t_k gives, for $1 \leq k \leq N$,

$$\frac{\mathbb{P}\{S \leq x\} - \Phi(x)}{L} \leq \sum_{j<k} C_j + \frac{\sum_{j\geq k} A_j + H + \Gamma_{T,t_k}(x)}{L}. \tag{9.2}$$

The first cell always uses the discrepancy bound C_0. An upper bound for Γ on $\mathbb{R}$ gives a bound uniform in x; we may take the minimum over any finite collection of cutoffs and smoothing parameters.

The reference comparison gives an additional bound depending on a closed threshold interval I. On a frequency cell $[l, r]$ let $[\alpha, \beta]$ enclose $\Re(k_T(t)e^{-\mathrm{i}xt})$ for all t in the cell and $x \in I$. If $|f_S - C_v| \leq E_j$ and $C_v - g \in [\ell_j, h_j]$, then

$$2\Re\int_l^r k_T(t)e^{-\mathrm{i}xt}(f_S(t) - g(t))\,dt \leq J_{j,I} := W_j E_j + 2(r-l)\max\{\ell_j\alpha, \ell_j\beta, h_j\alpha, h_j\beta\}, \tag{9.3}$$

where $W_j \geq 2\int_l^r |k_T|$. For a box $L \in [L_{lo}, L_{hi}]$, define

$$\mathcal{N}(z) = z/L_{lo} \quad (z \geq 0), \qquad \mathcal{N}(z) = z/L_{hi} \quad (z < 0).$$

It bounds division of the actual numerator by the actual L and is increasing in z. Hence an eligible cell uses $\min\{C_j, \mathcal{N}(J_{j,I})\}$; the first and all ineligible cells keep C_j.

The discrepancy and characteristic-function bounds can also be used together. Write $D = (f_S - g)/L$, and suppose $|D| \leq R$ and $|f_S| \leq r$. For arbitrary complex numbers k, h,

$$\Re(kD) = \Re((k-h)D) + \frac{\Re(hf_S) - g\Re h}{L} \leq R|k-h| + \frac{r|h| - g\Re h}{L}. \tag{9.4}$$

Use $k = k_T(t)e^{-\mathrm{i}xt}$ and choose a fixed rational complex h on each closed frequency–threshold rectangle. Interval evaluation of the right side, with $\mathcal{N}$ applied to its signed numerator, then gives a bound for the whole rectangle. Multiplying by twice its frequency width bounds its contribution to the integral. The choices $h = 0$ and $h = k$ explain the two individual modulus bounds; the calculation also uses intermediate values. Subdividing frequency or threshold intervals is valid when all child intervals are included.

Let $B_{j,I}$ denote the minimum of the discrepancy bound, the signed reference bound, and the applicable bounds from (9.4), each integrated over the full cell. The first cell uses $B_{0,I} = C_0$. Let c index the chosen smoothing parameters and their frequency grids, and let $\gamma_{c,k,I}$ be an upper bound for Γ on I at cutoff k. The resulting bound is

$$U_{c,k,I} = \sum_{j<k} B_{c,j,I} + \mathcal{N}\left(\sum_{j\geq k} A_{c,j} + H_c + \gamma_{c,k,I}\right).$$

For the finite partition of $[-X, X]$, the bound is

$$\max\left\{\max_I \min_{c,k} U_{c,k,I},\ \frac{1}{L_{lo}(1+X^2)}\right\}. \tag{9.5}$$

Cantelli bounds both exterior tails by $1/(1+X^2)$. Reflection gives the other Kolmogorov discrepancy: it preserves the moment box, real reference function and error bound. Its left-limit convention at atoms does not change the supremum against continuous Φ. The threshold partitions contain either 128 cells of width $1/32$ on $[-2,2]$, or 138 cells of that width on $[-69/32, 69/32]$. A 144-cell partition of the latter interval divides each of $[-15/32, -14/32]$ and $[-14/32, -13/32]$ into four equal parts. The exact endpoints and the chosen partition are recorded for each calculation. We also take the minimum with the bound uniform in x obtained from (9.2). For $X = 2$ the exterior numerator can be bounded by

$$\sup_{|x|\geq 2} |\mathbb{P}(S \leq x) - \Phi(x)| \leq \frac{1}{5} - (1 - \Phi(2)) < 0.17724986805182083. \tag{9.6}$$

Indeed, for $x \geq 2$ put $Q(x) = 1-\Phi(x)$ and $h(x) = (1+x^2)^{-1} - Q(x)$. Cantelli and reflection bound either Kolmogorov discrepancy by $\max\{Q(x), h(x)\}$. Mills' integral estimate $Q(x) \leq \phi(x)/x$ gives $2(1+x^2)Q(x) < 1$: the logarithmic derivative of $2(1+x^2)\phi(x)/x$ is $-(x^4+1)/(x(1+x^2)) < 0$, and its value at two is $5\phi(2) < 1$. Thus $h \geq Q$. Also $h'(x) = \phi(x) - 2x/(1+x^2)^2 < 0$, since the logarithmic derivative of $\phi(x)(1+x^2)^2/(2x)$ is $-(x^2-1)^2/(x(1+x^2)) \leq 0$ and its value at two is $25\phi(2)/4 < 1$. Both endpoint comparisons follow from $\phi(2) < e^{-2} < 1/7$. Hence $\max\{Q(x), h(x)\} \leq h(2)$. The estimate applies to open and closed tails. The stored evaluator parameters specify which exterior constant is used.

A negative upper product in (9.3) is multiplied by a lower bound for its positive width. Negative upper numerators use L_{hi}, and signed prefix sums round each addition upward. The nonnegative relative summation allowance cannot be applied to a negative sum.

9.1 Enclosing the Gaussian defect

Let $a = Ts$ and $h(u) = 1/(\pi u) - \cot(\pi u)$, with $h(0) = 0$. An exact signed expression is

$$\Gamma_{T,a}(x) = \int_0^s (1-u)e^{-T^2u^2/2}\{\cos(xTu) - h(u)\sin(xTu)\}\,du - \frac{1}{\pi}\int_s^\infty e^{-T^2u^2/2}\frac{\sin(xTu)}{u}\,du.$$

The ordinary Gaussian coefficient follows by taking absolute values inside both integrals; retaining the common threshold improves it.

The cotangent partial fractions give

$$p_m(u) = \sum_{j=0}^{m-1} \frac{2\zeta(2j+2)}{\pi}u^{2j+1}, \qquad 0 \leq h(u) - p_m(u) \leq \frac{\pi u^{2m+1}}{3(1-s^2)}.$$

Consequently $|\Gamma - \Gamma^{(m)}| \leq \epsilon_m$, where

$$\epsilon_m = \frac{\pi s^{2m+2}}{3(1-s^2)(2m+2)},$$

$$\Gamma^{(m)}(x) = -\tfrac{1}{2}\operatorname{erf}(x/\sqrt{2}) + \int_0^a g(t)\left[\frac{1-t/T}{T}\{\cos(xt) - p_m(t/T)\sin(xt)\} + \frac{x}{\pi}\operatorname{sinc}(xt)\right]dt.$$

The finite integrand is entire. Arb's implementation of the Petras integration method [5] supplies nodal integral enclosures; adding ϵ_m bounds the exact defect. At zero,

$$\Gamma_{T,a}(0) = \frac{\sqrt{\pi/2}}{T}\operatorname{erf}(a/\sqrt{2}) - \frac{1-e^{-a^2/2}}{T^2}.$$

For a global curvature bound put $H_s = \pi s/[3(1-s^2)]$. Differentiating twice and using $\sqrt{1+h^2} \le 1 + H_s$ gives

$$|\Gamma''(x)| \le M_G := (1+H_s)\left\{\frac{\sqrt{\pi/2}}{T}\operatorname{erf}(a/\sqrt{2}) - se^{-a^2/2}\right\} + \frac{e^{-a^2/2}}{\pi}.$$

If upper endpoint values on $[u,v]$ are G_u, G_v, then

$$\Gamma(x) \le \frac{(v-x)G_u + (x-u)G_v}{v-u} + \frac{M_G}{2}(x-u)(v-x). \tag{9.7}$$

Indeed the right side minus Γ is concave and nonnegative at both endpoints. Restricting this expression to intersections with a requested threshold cell gives its $\gamma_{c,k,I}$.

For the exterior thresholds, $Y > 0$ gives

$$\sup_{|x|\ge Y} \Gamma_{T,a}(x) \le \min\{1, 16/(T^2Y^2)\} + \operatorname{erfc}(Y/(2\sqrt{2})) + \frac{\sqrt{\pi/2}}{T}\operatorname{erfc}(a/\sqrt{2}) + \frac{E_1(a^2/2)}{2\pi}.$$

For a point mass at zero, let $\beta_T(y)$ be the signed Prawitz majorant minus the step function. Away from zero, $0 \le \beta_T(y)$ and $\beta_T(y) + \beta_T(-y) = \operatorname{sinc}^2(Ty/2)$. The full Gaussian defect is $\mathbb{E}\beta_T(x-Z)$ for a standard normal Z. Split at $|Z| \le Y/2$ to obtain the first two terms. Changing the full Gaussian cutoff to a costs at most the last two, using $2|K(u)| \le 1 + 1/(\pi u)$ and extending the positive integrals to infinity.

Finally the exact grid split sk/N need not equal the binary64 split $\widehat{s}$ stored in a Gaussian record. The uniform correction

$$|sk/N - \widehat{s}| \sup_{u\in I_s} 2|K(u)|,$$

where I_s joins them, follows by bounding the intervening signed Gaussian integral and using $g \le 1$. This correction is added before the coefficient is used in the matching cutoff. At the full cutoff, the analytic and Arb-integrated Gaussian coefficients may also be retained as valid upper bounds.

10 The interval calculation

The finite calculation covers $L \in [L_-, L_+]$, where the exact endpoints are

$$L_- = \frac{6456360425798343}{4611686018427387904} \approx 0.0014, \qquad L_+ = \frac{1362338887279575}{1125899906842624} \approx 1.21. \tag{10.1}$$

These are the rational values of the stored binary64 numbers.

10.1 Small and large moments

For $0 < L \le L_-$, Corollary 4.18 of Shevtsova [12] gives

$$\Delta/L \le C_{\mathrm{E}} + 0.3413L^{1/3} \le C_{\mathrm{E}} + 0.3413L_-^{1/3} < 0.447913037296. \tag{10.2}$$

The corollary applies to general independent arrays and requires $L \le 0.01$, which holds here.

For every mean-zero variance-one law, Cantelli's inequality and reflection give

$$\Delta \le \sup_{x>0}\left(\Phi(x) - \frac{x^2}{1+x^2}\right) < 0.540936541549. \tag{10.3}$$

At $x \geq 0$, Cantelli gives the lower CDF bound, while the opposite difference is at most $1 - \Phi(x) \leq 1/2$. Reflection handles $x < 0$, including one-sided limits at atoms. The displayed supremum exceeds $1/2$. Its stationary point is unique: the logarithmic derivative of $2x/((1+x^2)^2\phi(x))$ is $(x^2-1)^2/(x(1+x^2)) \geq 0$, and this ratio increases from zero to infinity. An Arb enclosure of the stationary value, divided by the exact L_+, gives

$$\Delta/L < 0.447055000001 \qquad (L \geq L_+). \tag{10.4}$$

10.2 Covering all feasible moments

For a closed interval $L \in [\ell, H]$, a containing box is

$$0 \leq d \leq \min\{1, H^{2/3}\}, \qquad 0 \leq b \leq H, \qquad 0 \leq \tau \leq \min\{1, H\}.$$

The constraints in (1.4) tighten this box, with outward rounding. Every remaining box is either accepted using a uniform upper bound, excluded by a necessary moment inequality, or divided into two closed children whose union contains its feasible part. A complete finite tree therefore bounds every array in its L interval. Different L intervals may overlap; exact rational endpoint comparisons check that their union has no gaps.

Some trees divide all three moment coordinates. Others divide (d, τ) and bound the entire feasible interval of b inside each such box. For fixed actual (L, d, τ), this interval is

$$d^{3/2} \leq b \leq L - \tau + d^{3/2}.$$

For a containing parameter box, outward endpoints give a containing b interval. If $U(I)$ bounds a slice with $b \in I$, then $\max_{I \in \mathcal{P}} U(I)$ bounds any complete closed partition $\mathcal{P}$ of that interval. During refinement, a parent is replaced only after both children have bounds; each child can also use the parent's bound. Thus stopping refinement at any stage retains coverage. For each box, the calculation evaluates a finite list of smoothing bounds and may stop once one gives the required upper bound. The stored evaluator parameters specify this stopping rule and hence which bound is returned.

Each smoothing calculation fixes $T > 0$ and $0 < s < 1$. Its stored binary64 parameters are interpreted as exact dyadic rationals. The meshes divide $[0, Ts]$ and $[Ts, T]$ into N intervals each, with the positive integer N specified in the record. The cutoffs in the first range are $Ts\,k/32$, $k = 24, \ldots, 32$. Parameter-dependent choices of T, s, and minima over several complete smoothing bounds, are permitted by the smoothing inequality. They impose no additional restriction on the moment domain.

A whole L interval may also be accepted without moment subdivision. The available bounds include (10.3) and the uniform smoothing estimates obtained from $\tau \leq L \leq H$ and $d \leq H^{2/3}$. In particular, taking $E_U = 2H$ and $d_U = \min(1, H^{2/3})$ gives

$$\mathcal{B}_U(t) = \min\{1, e^{-t^2/2 + \kappa E_U t^3 + m(t, d_U)}\}, \qquad F_U(t) = e^{-\psi(t; 1, E_U)}.$$

These follow from the product bounds and the classical zero-bias error.

Quantity	Recorded value
Closed L intervals	1,670
Partition nodes	483,030
Accepted leaves	235,659
Infeasible leaves	6,691
Largest upper bound for Δ/L	$\approx 0.44987999580838745$
Upper bound for $L \le L_-$	< 0.447913037296
Upper bound for $L \ge L_+$	< 0.447055000001

The counts include overlapping moment intervals. Among the accepted leaves, 143,618 have individually stored numerical values that were reproduced in separate evaluations with the same arithmetic routines. The other 92,041 leaves carry a valid upper bound for their entire containing L interval. A containing-interval bound is retained as such when checking an individual leaf. The table counts the outer moment trees. Some accepted leaves carry a further finite tree in b or d, with fixed numerical bounds on its closed subdomains and moment inequalities excluding its infeasible subdomains. These nested trees are checked in full; their nodes are not added to the outer counts. Each numerical record specifies the smoothing parameters, threshold partition, and applicable forcing bounds used in its evaluation. The largest finite upper endpoint, interpreted exactly, is

$$U_* = \frac{1013039690742047}{2251799813685248} < 0.44988. \tag{10.5}$$

Both analytic complements are smaller than U_*. The interval evaluations therefore give $\Delta/L \le U_*$ for every array. Taking the supremum gives the uniform bound $C_{\mathrm{ind}} \le U_* < 0.44988$.

10.3 Arithmetic

The vector calculations use correctly rounded IEEE-754 binary64 basic operations and square root, with gradual underflow and finite intermediate values. The supplied build uses no unsafe fast-math transformations. Endpoints are rounded outward with `nextafter`. For nonnegative sums the implementation uses an allowance depending on the mesh size, $1 + 4N2^{-53}$, including for every prefix of a cumulative sum. Signed sums use individually outward-rounded additions. Negative numerators use the upper endpoint of the positive normalizing interval, as described in the smoothing argument.

For a nonnegative exponent lower bound p, the vector code uses

$$P_{16}(x) = \sum_{j=0}^{16} \frac{x^j}{j!}, \qquad e^{-p} \le P_{16}(p/64)^{-64} \quad (0 \le p < 25).$$

It evaluates P_{16} downward, takes the reciprocal upward, and performs six upward-rounded squarings. At $p \ge 25$ it uses an upper enclosure of e^{-25}. Scalar transcendental values, geometric reductions and reference bounds use Arb ball arithmetic, at the precisions recorded with their sources. Gaussian node integrals use Arb integration with explicit analytic branches and remainder bounds [5]. The scalar majorant certificates give the coefficients and domains needed in the estimates above.

The supplement supplies the mathematical inputs and verification sources. `result.json` records the bound, counts, and evaluation methods; `verify/README.md` describes their scope and reproduction commands. The default checker reconstructs every closed parameter tree, verifies the necessary moment exclusions, compares the bounds as exact rational numbers, and checks the

union against the analytic endpoints. It also checks the stored Gaussian interpolation partitions and reconstructs each cutoff from its original smoothing parameter. The leaf integrals and Gaussian node integrals can be recomputed with the supplied arithmetic sources; the default check reads their saved enclosures. File hashes identify the precise mathematical inputs and routines.

The scalar inputs are in `certificates/scalar/`. The following map identifies the records and the functions in `verify/check_scalar_certificates.py` that check them.

Mathematical estimate	Certificate	Checking function
Logarithmic lower bounds, Lemmas 2.2 and 2.3	`direct-log.json`	`check_direct_log`
Quadratic imaginary-error bound, Lemma 5.1	`odd-quadratic.json`	`check_odd`
Affine bounds, (5.7)–(5.8)	`affine-index.json`, `disk.zip`, `directional.zip`	`check_affine`
Reference-error bound, (B.1)	`energy.json`	`check_energy`

The Gaussian interpolation records for (9.7) are in `certificates/gaussian-*.zip` and are checked by `verify/check_saved.py`. Together these archives contain the records for every cutoff used in the finite cover. The scalar guide, `certificates/scalar/README.md`, describes the record formats and separates checks of stored enclosures from their numerical reevaluation.

A The scalar majorant for the imaginary error

To verify (5.2), set $z = xy$, $q = 1 + x^2/3$, and

$$R(x,z) = \frac{\cos z + x^2 \operatorname{sinc} z - \cos x - x^2 \operatorname{sinc} x}{z^2 - x^2},$$

continued across $z = x$. Multiplication of the difference in (5.2) by $x(z + x/2)$ gives

$$\mathcal{L}(x,z) = x(z+x/2)\left\{J_x(1) + \frac{J_x'(1)}{2x^2}(z^2 - x^2)\right\} + \frac{(z-x)^2(z+x/2)^2}{4} - z^2(z+x)^2 R(x,z)^2.$$

It has a double zero at $z = x$. Define

$$w_n(u) = \frac{1}{(2n)!} + \frac{u}{(2n+1)!}, \qquad S_n(x,z) = \sum_{j=0}^{n-1} x^{2j} z^{2(n-1-j)}.$$

For $r = m + n$, the polynomial

$$Q_{mn}(x,z) = \frac{z^2(z+x)^2 S_m S_n - \frac{4mn}{9} x^{2r-3}(z+x/2)\{(5-3r)x^2 + (3r+1)z^2\}}{(z-x)^2}$$

is well defined by exact division. Expansion and collection of terms give

$$\frac{\mathcal{L}(x,z)}{(z-x)^2} = \frac{q^2 x(x+4z)}{144} - \frac{(q^2-1)(z+x/2)^2}{4} + \sum_{r\ge 3}(-1)^{r+1} \sum_{m=1}^{r-1} w_m(x^2) w_{r-m}(x^2) Q_{m,r-m}(x,z). \tag{A.1}$$

This identity also defines the removable diagonal. Each Q_{mn} is homogeneous of degree $2r-2$, has positive rational coefficients, and $Q_{mn}(1,1) \le 8mnr^2$. One way to check these assertions is to put $x = 1$. The polynomial numerator is

$$N(z) - M(z) = z^2(z+1)^2 S_m(1,z) S_n(1,z) - \frac{4mn}{9}(z+1/2)\{5 - 3r + (3r+1)z^2\}.$$

The constant and linear coefficients of its quotient by $(z-1)^2$ are $2mn(3r-5)/9$ and four times that value. Higher coefficients are positive sums of coefficients of N. Also $Q_{mn}(1,1) = \{N''(1) - M''(1)\}/2 \le 8mnr^2$.

Truncate (A.1) at $r = 24$ and write its polynomial uniquely as

$$x^2 F_1(x^2, z^2) + z^4 F_0(z^2) + xz G(x^2, z^2),$$

assigning positive even powers of x to the first term. On $(x^2, z^2) \in [0,4] \times [0,49]$, exact tensor Bernstein conversion gives minimum coefficients $1/144$ for F_1, greater than $1/200$ for F_0, and $1/36$ for G. These inequalities follow from (A.1) by the accompanying rational checker; its scalar certificate records the exact minima. The conversion itself uses, after scaling to $[0,1]^2$,

$$b_{ij} = \sum_{p \le i,\, q \le j} a_{pq} \frac{\binom{i}{p}}{\binom{d_1}{p}} \frac{\binom{j}{q}}{\binom{d_2}{q}}.$$

Bernstein basis functions are nonnegative and sum to one.

To control the omitted series, use $w_n(x^2) \le 7/[3(2n)!]$ and

$$\sum_{m=1}^{r-1} \frac{m(r-m)}{(2m)!(2r-2m)!} = \frac{2^{2r-5}}{(2r-2)!}.$$

The positive order-r term at $(x,z) = (2,7)$ is at most

$$B_r = \frac{49}{1764}\frac{r^2 196^r}{(2r-2)!}.$$

It equals $4F_{1,r} + 2401F_{0,r} + 14G_r$, so each component is at most $B_r/4$. For $r \ge 25$, $B_{r+1}/B_r < 1/10$; each absolute component tail is therefore at most $5B_{25}/18 < 1/1250$. The rational checker verifies that each minimum minus this tail exceeds $1/250$. Thus $\mathcal{L}/(z-x)^2 \ge (x^2 + z^4 + xz)/250 \ge 0$ on the rectangle.

For $z \ge 7$, the elementary bound $|A_x(y) - A_x(1)| \le 2(x + 1/x)$ gives

$$\frac{J_x(y)}{(x^2/4)h(y)} \le \frac{16(1+x^2)^2 z^2}{(z-x)^4(z+x/2)^2} \le \frac{400}{(z-2)^4} \le \frac{16}{25} < 1.$$

The two tangent terms in (5.2) are nonnegative there. Indeed $J_x(1) \ge 0$, while $J_x'(1) > 0$: writing $k_0 = 2\operatorname{sinc} x - \cos x$ and $\ell_0 = (2-x^2)\operatorname{sinc} x - 3\cos x$ gives $J_x'(1) = 2x^2 k_0(4k_0 - 3\ell_0)/9$; $k_0 > 0$, and alternating Taylor bounds give $4k_0 - 3\ell_0 \ge 257/105$ on $[0,2]$. This proves the remaining spatial tail and completes (5.2).

B The scalar bound for the cosine comparison

To prove (7.4), for $x > 0$ set

$$a_x = \frac{2x - x\sec x - \sin x}{x^2}, \quad U_x(w) = T_x(w) - a_x(w^2-1), \quad V_x(y) = \frac{x^2\cos x}{2} U_x(y).$$

Writing $s = \sin x$, $c = \cos x$, $A = (x+sc)/2$, $B = s^2$ and $D = x^2 + 2xsc$, one has

$$V_x(y) = (\mathrm{i}yc + s)e^{\mathrm{i}xy} - \mathrm{i}y + A(y^2-1).$$

Set $q_v = 2B^2/3$, $r_v = 2BD/3 - 4B^2/9$, $\lambda_v = \lambda x^4 c^2/4$, with $\lambda = 9/8$. We verify the scalar inequality

$$\mathcal{D}_x(y) := h(y)\{q_v + (r_v/2)(y^2-1) + \lambda_v h(y)\} - |V_x(y)|^2 \ge 0 \qquad (y \ge 0). \tag{B.1}$$

The value and first three derivatives at $y = 1$ vanish. This exact cancellation is removed before interval evaluation.

Define

$$E_n(\mathrm{i}v) = \sum_{m \ge 0} \frac{(\mathrm{i}v)^m}{(m+n)!} = \frac{1}{(n-1)!}\int_0^1 (1-u)^{n-1} e^{\mathrm{i}vu}\, du.$$

With $z = y - 1$, the regular quotient is

$$\frac{\mathcal{D}_x(y)}{(y-1)^4} = \lambda_v(y+1/2)^2 + r_v(y/2 + 5/4) - A^2 - \Re \sum_{n=1}^{4} c_n E_n(\mathrm{i}xz),$$

where

$$c_4 = -2x^4, \qquad c_3 = 2x^4 + 2\mathrm{i}x^3(1+c^2),$$
$$c_2 = x^2(4Asc + 2c^2) - 2\mathrm{i}x^2\{A(1+2c^2) - sc\}, \qquad c_1 = -2xAc^2 - 2\mathrm{i}xAsc.$$

These coefficients follow by expanding $|V_x|^2$ and subtracting its Taylor polynomial of degree three at $y = 1$. The degree-K Taylor polynomial for $E_n(\mathrm{i}v)$ has error at most $\max|v|^{K+1}/(n+K+1)!$, by its integral representation.

For cells touching $x = 0$, remove the factor x^4 as well. Put $S = \operatorname{sinc} x$, $u = \operatorname{sinc}(2x) = Sc$, $a = (1+u)/2$,

$$R = \tfrac{2}{3}S^2(1+2u) - \tfrac{4}{9}S^4, \qquad K_0 = -uS^2 - 4x^2[\Re E_3(2\mathrm{i}x)]^2.$$

Then the analytic quotient $\mathcal{H} = \mathcal{D}/[x^4(y-1)^4]$ is

$$\mathcal{H}(x,y) = \tfrac{\lambda}{4}c^2(y+1/2)^2 + R(y/2+5/4) + K_0 - 2au(z+2)\Re E_2(\mathrm{i}xz)$$
$$- 2\{ac^2z^2 + [a(1+2c^2) - u]z + 1\}\Re E_3(\mathrm{i}xz) + 2\{c^2z^2 + (1+c^2)z + 1\}\Re E_4(\mathrm{i}xz). \quad \text{(B.2)}$$

For example, $\Re E_1(\mathrm{i}v) = 1 - v^2\Re E_3(\mathrm{i}v)$ and $\Im E_n(\mathrm{i}v) = v\Re E_{n+1}(\mathrm{i}v)$ for $n = 1, 2, 3$ give this formula from the preceding quotient. The remaining constant cancellation is

$$(uc^2 - a^2)/x^2 = -uS^2 - (1-u)^2/(4x^2) = -uS^2 - 4x^2[\Re E_3(2\mathrm{i}x)]^2.$$

At zero,

$$\mathcal{H}(0,y) = (4y+1)/144 + (\lambda - 1)(y+1/2)^2/4 > 0.$$

The recorded scalar partitions cover $0 \le x \le 1/2$, $0 \le y \le 48$ using (B.2), and $1/2 \le x \le 1$, $0 \le y \le 16$ using the positive-frequency quotient. Every accepted rectangle has an outward nonnegative lower bound; all rectangle joins are exact. There are analytic infinite tails. On $0 \le x \le 1/2$,

$$|U_x(y)| \le P(y) := y^3 + \tfrac{20}{21}y^2 + \tfrac{8}{7}y + \tfrac{4}{21}.$$

For $x > 0$ this follows by writing $V = (\mathrm{i}yc + s)(e^{\mathrm{i}xy} - 1 - \mathrm{i}xy) + (A - xc)y^2 + s - A + \mathrm{i}y(c - 1 + xs)$, using $c \ge 7/8$, $|e^{\mathrm{i}v} - 1 - \mathrm{i}v| \le v^2/2$, $0 \le A - xc \le x^3/3$, $0 \le s - A \le x^3/6$, and $0 \le c - 1 + xs \le x^2/2$. At zero it follows directly. For $y \ge 1$, the unscaled gap is at least $\lambda h(y)^2 - P(y)^2$. At $y = 48 + z$ its exact coefficients, in increasing order, are

$$\frac{51794446417}{288}, \quad \frac{354649842}{7}, \quad \frac{28964396957}{7056}, \quad \frac{181934075}{1176}, \quad \frac{43072585}{14112}, \quad \frac{5161}{168}, \quad \frac{1}{8}.$$

They are positive. On $1/2 \le x \le 1$, use $A \le 3/4$ and $x^4\cos^2 x/4 > 1/100$ to obtain

$$\mathcal{D}_x(y) \ge \frac{1}{100}(y-1)^4(y+1/2)^2 - (3y^2/4 + 2y + 1)^2.$$

At $y = 16 + z$ the coefficients are

$$\frac{1395225}{16}, \quad 41760, \quad \frac{60985}{8}, \quad \frac{14097}{20}, \quad \frac{1773}{50}, \quad \frac{93}{100}, \quad \frac{1}{100},$$

again positive. These tails and the compact certificates prove (B.1) for all $y \ge 0$; parity supplies real w.

Finally, divide the unscaled gap by h at its continuous value at one. Weighted Cauchy–Schwarz gives

$$|\mathbb{E}U_x(W)|^2 \le \mathbb{E}h(Y)\,\mathbb{E}\frac{|U_x(W)|^2}{h(Y)} \le (\rho - 1)\{q_{\mathrm{ref}}(x) + \lambda(\rho - 1)\}.$$

Since $\mathbb{E}U_x = \mathbb{E}T_x$, the scalar enclosures and the analytic tails prove (7.4). The integrand $|U_x(w)|^2/h(|w|)$ grows at most linearly at positive x and cubically at $x = 0$, so the third absolute moment suffices. If $\rho = 1$, $|W| = 1$ almost surely and the forcing is zero.

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